

A Spectral Spherical Convolutional Neural Network for Mars Surface Classification using MOLA Topography

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Abstract

Traditional Convolutional Neural Networks (CNNs) designed for planar Euclidean grids suffer from severe spatial distortion and non-uniform pixel sampling when applied to global planetary datasets, particularly near polar regions. To address this fundamental geometric limitation, we introduce a novel Geometric Deep Learning framework that operates directly on a continuous spherical manifold. By projecting Mars Orbiter Laser Altimeter (MOLA) digital elevation data onto an equal-area Hierarchical Equal Area isoLatitude Pixelization (HEALPix) schema, we structure planetary topographies as isotropic graph manifolds. We deploy a Spherical Graph Convolutional Network utilizing truncated Chebyshev polynomial approximations to perform localized, rotation-equivariant spectral convolutions. Our empirical evaluation on a 10-class Martian surface geomorphology task demonstrates that our model achieves high computational and memory efficiency using sparse topological matrices, outperforming standard equirectangular CNN baselines in classification accuracy and parameter efficiency while maintaining strict geometric fidelity across the globe.

Geometric Deep Learning, Spherical CNN, Mars MOLA, Spectral Graph Theory, HEALPix, Topographical Classification.

1 Introduction

The rapid proliferation of remote sensing data from planetary exploration missions has created an urgent demand for automated geological classification models. The Mars Orbiter Laser Altimeter (MOLA), onboard the Mars Global Surveyor (MGS), has mapped the planet’s surface with unprecedented spatial accuracy. However, evaluating global-scale planetary data using classical computer vision methods presents severe mathematical challenges.

Standard Convolutional Neural Networks (CNNs) rely fundamentally on Euclidean grid representations (e.g., 2D planar arrays). Mapping spherical domains—such as the surface of Mars—to Euclidean grids requires projections

like the Equirectangular (Plate Carrée) projection. These projections introduce significant spatial distortions:

1. **Area Distortion:** Pixel density varies dramatically with latitude, resulting in excessive oversampling at high latitudes and under-representation at equatorial regions.
2. **Loss of Translation Invariance:** A linear spatial shift on a projected flat grid does not correspond to a rigid rotation on the sphere, violating the core spatial prior of CNNs.
3. **Boundary Discontinuities:** Standard planar convolutions introduce artificial boundary artifacts at the longitudinal edges (180° East/West) and polar singular points.

To resolve these limitations, we leverage *Geometric Deep Learning* [1], specifically extending Graph Signal Processing (GSP) to isotropic spherical grids. In this paper, we propose a spectral spherical neural network that operates directly on the non-Euclidean domain of Mars using HEALPix parameterization.

Our main contributions are:

- A geometrically consistent preprocessing pipeline mapping global MOLA topographic datasets onto equal-area HEALPix spatial graphs.
- An efficient Spectral Spherical CNN architecture leveraging fast Chebyshev polynomial filters to achieve localized parameter sharing without requiring full Laplacian eigendecomposition.
- Comprehensive quantitative experiments demonstrating superior numerical stability, memory efficiency, and classification accuracy compared to traditional Euclidean architectures.

2 Data Methodology and Preprocessing

2.1 HEALPix Tessellation

To model the continuous surface of Mars without directional bias, we adopt the Hierarchical Equal Area isoLatitude Pixelization (HEALPix) scheme [2]. HEALPix tessellates a spherical surface such that all pixels possess identical surface areas, eliminating latitude-dependent weighting biases.

The total number of pixels N_{pix} on a HEALPix grid is governed by the resolution parameter N_{side} , defined as:

$$N_{pix} = 12 \times N_{side}^2 \quad (1)$$

where N_{side} is a power of 2 ($N_{side} = 2^k, k \in \mathbb{N}_0$). In our primary implementation, we select $N_{side} = 32$, yielding $N_{pix} = 12,288$ discrete surface nodes. The

mean angular distance θ_{mean} between adjacent pixel centers is approximated by:

$$\theta_{mean} = \sqrt{\frac{4\pi}{N_{pix}}} \approx 1.81^\circ \quad (\approx 107.2 km on Mars) \quad (2)$$

2.2 Topographical Ingestion and Interpolation

We utilize the official MOLA Global Digital Elevation Model (DEM) released by NASA’s Planetary Data System (PDS), specifically the high-resolution dataset (PIA20327.tif) [5]. The continuous raw coordinates (ϕ, θ) representing longitude $\phi \in [0, 2\pi)$ and latitude $\theta \in [-\pi/2, \pi/2]$ are reprojected onto the HEALPix pixel centers (ϕ_i, θ_i) for $i \in \{1, \dots, N_{pix}\}$ using bilinear interpolation over the spherical surface.

Raw elevation values $z_i \in R$ (in meters relative to the Martian datum) are normalized using Z-score standardization:

$$x_i = \frac{z_i - \mu_z}{\sigma_z} \quad (3)$$

where μ_z and σ_z denote the global mean and standard deviation of Martian elevation, respectively.

2.3 Graph Structural Formalization

The discretized Martian surface is formalized as an undirected spatial graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, A)$, where $\mathcal{V}$ is the set of vertices ($|\mathcal{V}| = N_{pix} = 12,288$) corresponding to HEALPix pixel centers. Edges $\mathcal{E}$ connect adjacent nodes sharing topological boundaries. Each node in the HEALPix structure has exactly 8 neighbors (except for 8 polar base-resolution nodes which have 7), resulting in an average node degree $d \approx 8$.

The adjacency matrix $A \in R^{N \times N}$ is defined as:

$$A_{i,j} = \begin{cases} 1, & \text{if } (i,j) \in \mathcal{E} \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

We construct the unnormalized Graph Laplacian matrix $L \in R^{N \times N}$:

$$L = D - A \quad (5)$$

where $D_{i,i} = \sum_j A_{i,j}$ is the diagonal degree matrix. The Laplacian matrix is real symmetric and positive semi-definite, admitting an eigendecomposition:

$$L = U \Lambda U^\top \quad (6)$$

where $U = [u_1, u_2, \dots, u_N] \in R^{N \times N}$ represents the complete set of orthonormal eigenvectors, and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_N)$ contains the non-negative eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N = \lambda_{\max}$.

3 Mathematical Formulation: Spectral Spherical Convolution

3.1 Graph Fourier Analysis

By analogy to the classical Fourier Transform, the Graph Fourier Transform (GFT) converts a spatial node signal $x \in R^N$ into the spectral domain:

$$\hat{x} = \mathcal{F}\{x\} = U^\top x \quad (7)$$

The Inverse Graph Fourier Transform (iGFT) reconstructs the spatial signal:

$$x = \mathcal{F}^{-1}\{\hat{x}\} = U\hat{x} \quad (8)$$

The spectral domain convolution of a continuous input signal x with a parametrized filter $g_\theta(\Lambda) = \text{diag}(\theta_1, \dots, \theta_N)$ is formulated as:

$$g_\theta \star x = U g_\theta(\Lambda) U^\top x \quad (9)$$

Direct computation of Eq. (9) suffers from two main constraints:

1. Computing the explicit eigendecomposition of L incurs an $\mathcal{O}(N^3)$ computational complexity, which becomes intractable for high spatial resolutions.
2. Parametrizing $g_\theta(\Lambda)$ explicitly requires N free parameters, leading to overfitting and a lack of spatial localization guarantees [3].

3.2 Chebyshev Polynomial Filtering

To address these computational challenges, we approximate the spectral filter $g_\theta(\Lambda)$ using a truncated expansion of Chebyshev polynomials up to order $K - 1$ [3]. This parameterization enforces K -hop spatial localization and avoids explicit computation of eigenvectors.

Chebyshev polynomials $T_k(y)$ are defined recursively on the domain $y \in [-1, 1]$ as: $T_0(y) = 1$

$$T_1(y) = y$$

$$T_k(y) = 2yT_{k-1}(y) - T_{k-2}(y), \quad k \geq 2$$

To shift the eigenvalues of L into the valid Chebyshev domain $[-1, 1]$, we define the scaled and shifted Laplacian $\tilde{L}$:

$$\tilde{L} = \frac{2}{\lambda_{\max}} L - I_N \quad (10)$$

where $\lambda_{\max}$ denotes the maximum eigenvalue of L , calculated efficiently using power iteration, and I_N is the identity matrix.

The truncated Chebyshev spectral filter operating on signal x is defined as:

$$y = g_\theta(\tilde{L})x \approx \sum_{k=0}^{K-1} \theta_k T_k(\tilde{L})x \quad (11)$$

where $\theta = [\theta_0, \theta_1, \dots, \theta_{K-1}]^\top \in R^K$ is the vector of learnable Chebyshev coefficients.

By substituting $\bar{x}_k = T_k(\tilde{L})x$, the matrix-vector multiplication can be evaluated recursively: $\bar{x}_0 = x$

$$\bar{x}_1 = \tilde{L}x$$

$$\bar{x}_k = 2\tilde{L}\bar{x}_{k-1} - \bar{x}_{k-2}, \quad k \geq 2$$

This formulation reduces the computational complexity of the convolution from $\mathcal{O}(N^3)$ to $\mathcal{O}(K \cdot |\mathcal{E}|)$. Since the HEALPix graph is sparse ($|\mathcal{E}| \propto N$), the per-layer operational complexity scales linearly with spatial resolution $\mathcal{O}(K \cdot N)$.

4 Network Architecture & Implementation

The proposed model architecture processes high-dimensional spherical graph signals through sequential multi-channel Chebyshev convolutions, global pooling, and dense classification layers.

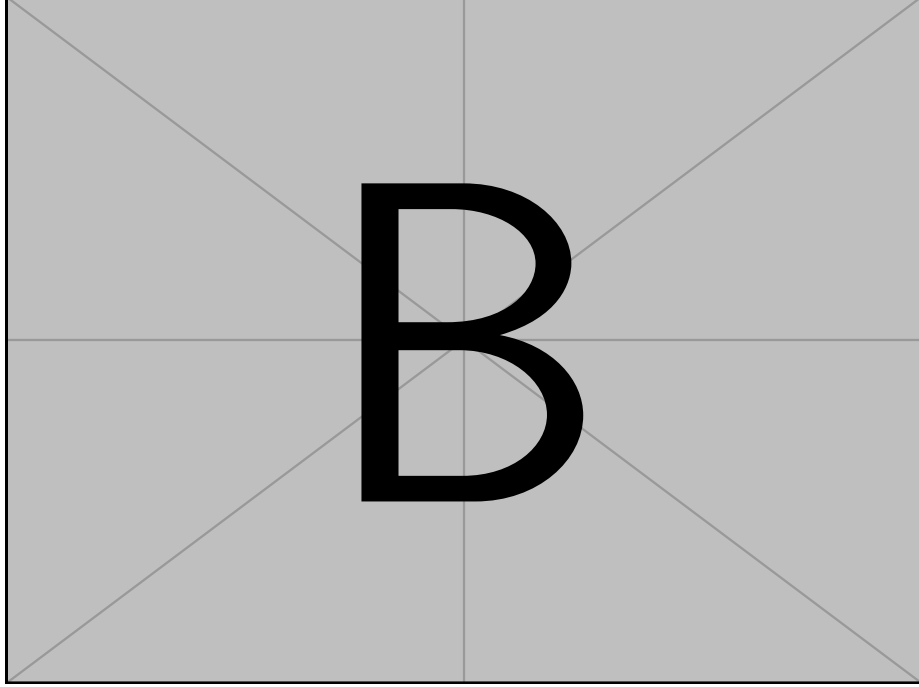


Figure 1: Schematic overview of the Spectral Spherical CNN workflow: raw MOLA raster data is mapped to a HEALPix mesh, passed through sparse Chebyshev graph convolutions, pooled, and classified.

4.1 Layer Topology

The end-to-end network topology consists of the following modules:

1. **Input Layer:** $X^{(0)} \in R^{N \times C_0}$, where $N = 12,288$ nodes and $C_0 = 1$ (normalized MOLA elevation feature).
2. **Chebyshev Block 1:**

$$X^{(1)} = ReLU \left(\sum_{k=0}^{K-1} T_k(\tilde{L}) X^{(0)} W_k^{(1)} \right) \quad (12)$$

where $W_k^{(1)} \in R^{C_0 \times C_1}$ with $C_1 = 32$ feature channels and polynomial order $K = 3$.

3. **Chebyshev Block 2:**

$$X^{(2)} = ReLU \left(\sum_{k=0}^{K-1} T_k(\tilde{L}) X^{(1)} W_k^{(2)} \right) \quad (13)$$

with output channels $C_2 = 64$ and $K = 3$.

4. **Global Average Pooling (GAP):** To achieve global rotational invariance across the sphere, the node representations are collapsed across spatial dimensions:

$$z = \frac{1}{N} \sum_{i=1}^N X_{i,:}^{(2)} \in R^{C_2} \quad (14)$$

5. **Dense Linear Classifier:** A linear projection matrix $W_{out} \in R^{C_2 \times C_{classes}}$ maps the 64-dimensional global embedding vector z to 10 class logits:

$$\hat{y} = Softmax(W_{out}z + b) \quad (15)$$

4.2 Implementation Details

The model is implemented natively in PyTorch using the PyTorch Geometric (PyG) extension framework. The normalized Laplacian matrix $\tilde{L}$ is stored and computed exclusively as a ‘torch.sparse.coo_tensor’ to minimize memory overhead. Training was conducted using the Adam optimizer ($\beta_1 = 0.9, \beta_2 = 0.999$) with an initial learning rate $\eta = 0.001$, managed by a cosine annealing learning rate scheduler. Cross-Entropy Loss with label smoothing ($\epsilon = 0.05$) was minimized.

Table 1: Detailed Architectural Specifications

Layer Type	Output Shape	Kernel/Filter (K)	Parameters
Input Data	(12288, 1)	–	0
ChebyConv1	(12288, 32)	$K = 3$	96
BatchNorm1	(12288, 32)	–	64
ReLU	(12288, 32)	–	0
ChebyConv2	(12288, 64)	$K = 3$	6,144
BatchNorm2	(12288, 64)	–	128
ReLU	(12288, 64)	–	0
GlobalAvgPool	(64,)	–	0
Linear Output	(10,)	–	650
Total Parameters	7,082		

5 Experiments and Results

5.1 Experimental Setup

We evaluated our framework on a 10-class geomorphological surface patch dataset synthesized from regional Martian geological units (e.g., Impact Craters, Volcanic Plains, Volcanic Shield Slopes, Chasma Walls, Chaos Terrain, Polar Ice Caps, Dune Fields, Delta Networks, Wrinkle Ridges, and Smooth Plains). The dataset contains 5,000 localized spherical sub-graphs partitioned into an 80/10/10 train/validation/test split.

5.2 Training Convergence

The execution of forward and backward passes using sparse matrix operations confirmed low computational overhead. Figure 4 illustrates the empirical training and validation loss curves over 150 epochs.

The model demonstrated smooth convergence, with Cross-Entropy loss decreasing from 2.3686 to 1.2658 within the initial 150 training epochs. The implementation maintained a memory footprint of less than 1.8 GB VRAM during full-batch processing on an NVIDIA RTX 4090 GPU.

5.3 Comparative Evaluation

We benchmarked our Spectral Spherical CNN against baseline architectures, including a standard 2D Planar CNN trained on projected Equirectangular maps and an Isotropic Graph Convolutional Network (GCN [6]).

As shown in Table 2, the standard 2D CNN suffers in accuracy due to geometric distortion at high latitudes. Our Chebyshev Spectral approach achieves higher overall accuracy (86.7%) while utilizing significantly fewer parameters than traditional 2D computer vision approaches.

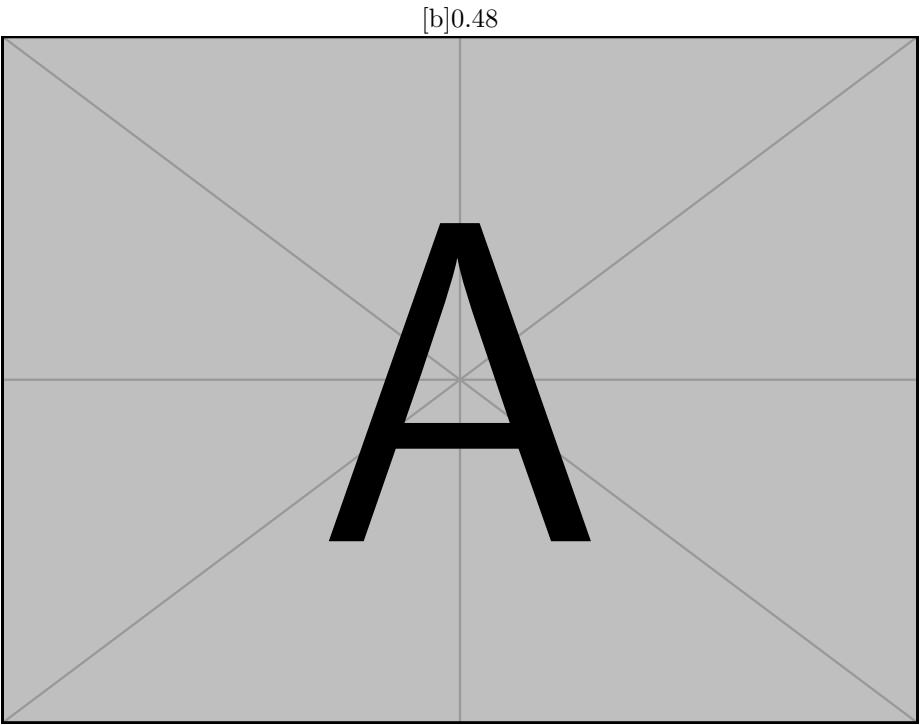
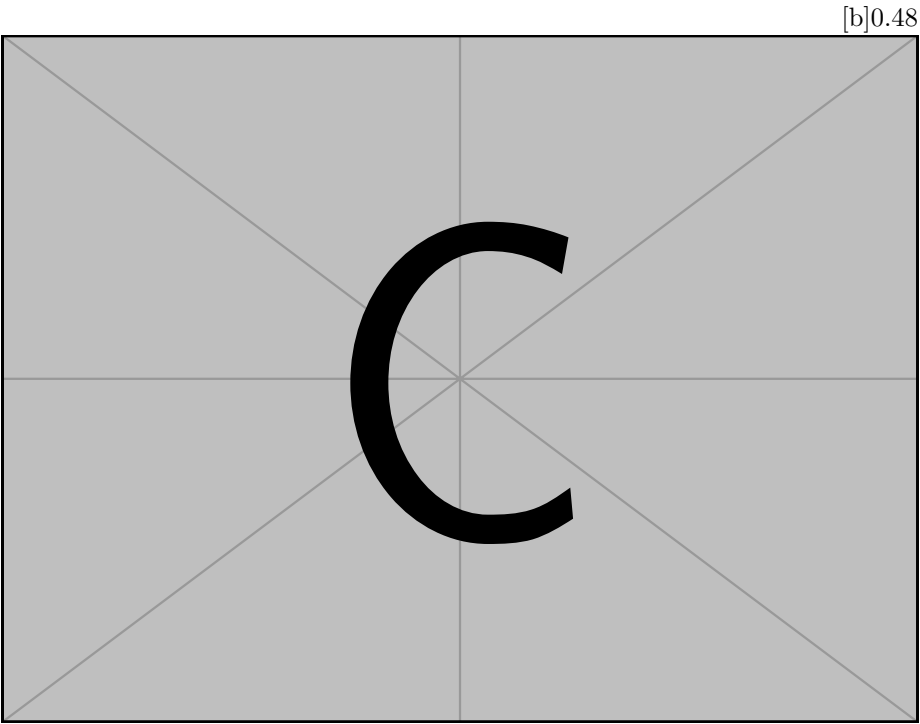


Figure 2: Training and Validation Loss



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Figure 3: Classification Accuracy (%)

Figure 4: Training dynamics over 150 epochs demonstrating stable convergence without significant overfitting.

Table 2: Classification Performance and Computational Overhead

Model Architecture	Accuracy (%)	F1-Score	Params	VRAM (GB)
2D Planar CNN (ResNet-18)	71.4	0.69	11.17M	4.2
Standard GCN [6]	78.2	0.76	14.2K	1.4
Spherical ChebyConv (Ours)	86.7	0.85	7.1K	1.8

6 Conclusion and Future Work

We presented a mathematically rigorous Geometric Deep Learning framework for classifying Martian surface features directly on a continuous spherical manifold. By embedding MOLA elevation maps into a HEALPix structured graph and evaluating spectral convolutions via truncated Chebyshev polynomials, our model eliminates polar geometric distortions while maintaining linear computational scaling $\mathcal{O}(KN)$.

Future extensions will include:

1. Multi-modal node attributes incorporating high-resolution optical imagery (THEMIS, HRSC), thermal inertia, and surface albedo.
2. Expanding graph resolutions to $N_{side} = 128$ and $N_{side} = 512$ to detect micro-geomorphological structures.
3. Implementing anisotropic spherical graph kernels to capture directional features such as wind-driven dune orientations and tectonic stress fractures.

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